

Dilan Karaguler

Ph.D. Candidate in Mathematics, Michigan State University

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RESEARCH INTERESTS

Applied algebraic topology and topological data analysis; path homology and Mayer (N-nilpotent) homology of directed graphs and path complexes; persistent homology; topology-informed graph machine learning and geometric deep learning, with applications to molecular, dynamical and engineering data.

EDUCATION

Michigan State University

Ph.D. in Mathematics (Candidate)

– Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

East Lansing, MI, US

August 2022 – May 2027 (Expected)

Middle East Technical University (METU)

M.S. in Mathematics

– Thesis: *A Survey on Multidimensional Persistence Theory*

– Advisor: Assoc. Prof. Semra Pamuk

Ankara, Türkiye

September 2019 – September 2021

Middle East Technical University (METU)

B.S. in Mathematics, Honor Student

Ankara, Türkiye

June 2014 – June 2019

PUBLICATIONS

D. Karaguler and G.-W. Wei. *Mayer Path Homology*. Foundations of Data Science, accepted (2026).
doi:10.3934/fods.2026025, [arXiv:2605.16525](https://arxiv.org/abs/2605.16525).

D. Karaguler. *A Survey on Multidimensional Persistence Theory*. M.S. Thesis, Middle East Technical University (2021).

D. Karaguler and G.-W. Wei. *Persistent Mayer Path Laplacians*. In preparation (2026).

RESEARCH EXPERIENCE

Graduate Researcher

Michigan State University

- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

August 2022 – Present

East Lansing, MI, US

Research Mentor

Topology RTG, Michigan State University

- Mentored an undergraduate researcher on a topological data analysis project: project design, methodology and research presentation.

May 2023 – June 2023

East Lansing, MI, US

INDUSTRY EXPERIENCE

Software Engineer Intern

Intel Corporation

- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

May 2026 – August 2026

Oregon, US

Graduate Technical Intern

Intel Corporation

- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

May 2025 – August 2025

Oregon, US

TEACHING EXPERIENCE

Lead Teaching Assistant

August 2023 – Present

Department of Mathematics, Michigan State University

East Lansing, MI, US

- Lead and mentor a team of graduate and undergraduate teaching assistants; observe in-person and online sections and give constructive feedback.
- Currently lead TA for first-year graduate students; design and run workshops that prepare them for their first teaching assignments.
- Work with faculty to set best practices for teaching assistants and prepare teaching materials for multi-section courses.

Graduate Lead, Center for Instructional Mentoring (CIM)

Summer 2024

Department of Mathematics, Michigan State University

East Lansing, MI, US

- Mentored first-time instructors through non-evaluative teaching feedback, one-on-one conversations and ongoing guidance during their first teaching experience.
- Created a teaching handbook built on interviews with senior TAs about the courses they taught, describing each course's expectations so junior graduate students can choose assignments that fit their needs.

Mentor, Academic Culture and English Skills (ACES) Program

Summer 2024

Michigan State University

East Lansing, MI, US

- Mentored incoming international graduate students preparing to become MSU teaching assistants, introducing U.S. classroom culture, university structure and cross-cultural communication, and organized social events.

Supervisor, Math Learning Center

August 2022 – May 2023

Michigan State University (Teaching Exemption Fellowship)

East Lansing, MI, US

Graduate Teaching Assistant

January 2020 – June 2022

Middle East Technical University

Ankara, Türkiye

- Led recitations and office hours, graded, and proctored online and in-person exams for Calculus I–II and Differential Equations.

Undergraduate Teaching Assistant

March 2018 – June 2019

Middle East Technical University

Ankara, Türkiye

- Graded homework and held review hours for Calculus for Mathematics Students.

COURSES TAUGHT

MSU MTH 299 — Transitions	Fall 2025
MSU MTH 314 — Matrix Algebra with Computational Applications	Spring 2025
MSU MTH 235 — Differential Equations	Spring 2024
MSU MTH 133 — Calculus II	Summer 2023, Fall 2023, Fall 2024
METU MTH 154 — Calculus II for Mathematics Students	Spring 2022
METU MTH 153 — Calculus I for Mathematics Students	Fall 2021
METU MTH 120 — Calculus II for Engineering Students	Spring 2021
METU MTH 119 — Calculus I for Engineering Students	Fall 2020

TEACHING DEVELOPMENT & EDUCATIONAL RESEARCH

Program Evaluator & Instructional Researcher

May 2026

AI for Advanced Semiconductor Manufacturing Workshop, Michigan State University

East Lansing, MI, US

- Designed pre/post-workshop surveys and observation protocols to evaluate program effectiveness and adult learning outcomes.
- Analyzed participant feedback quantitatively and with qualitative thematic coding, tracking confidence growth across AI and semiconductor manufacturing topics.
- Wrote an evaluation report grounded in adult learning theory with curriculum recommendations for the workshop.

Colleges Online Learning Academy (COLA) Scholar

Summer 2024

Michigan State University

Portfolio

- Built a teaching portfolio website and wrote a project plan for online course design as program deliverables.

TALKS

Mayer Path Homology

TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU

Mayer Homology and Applications

Invited lecture, AI in Mathematics course, Michigan State University

Intersection Homology and Applications

Invited lecture, AI in Mathematics course, Michigan State University

From Data to Barcodes

Graduate Student Seminar, Middle East Technical University

April 2021

Ankara, Türkiye

POSTER PRESENTATIONS

SIAM Great Lakes Section Meeting

Illinois Institute of Technology

September 2025

Chicago, IL, US

Mathematics of Data, Dynamics, and Life Sciences Conference

University of California, Irvine

March 2025

Irvine, CA, US

PROJECTS

Persistent Mayer Homology with Dynamic Mode Decomposition

Python, NumPy, SciPy, GUDHI

April 2026

[GitHub](#)

- Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
- Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

Homology-Fed Neural Network Framework

Python, PyTorch, GUDHI, NumPy

November 2024

[GitHub](#)

- Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
- Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

Cellular Automata on Curved Surfaces

Python, NumPy, Matplotlib

March 2024

[GitHub](#)

- Simulated Conway’s Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

AWARDS, FELLOWSHIPS & SCHOLARSHIPS

Excellence in Teaching Award, Department of Mathematics, Michigan State University

May 2025

Colleges Online Learning Academy (COLA) Fellowship, Michigan State University

Summer 2024

Teaching Exemption Fellowship, Michigan State University

2022 – 2023

RONESANS Scholarship, RONESANS

2018 – 2019

UNIGENC Scholarship, UNIGENC Association

2015 – 2019

Merit Scholarship, Turkish Ministry of National Education

2010 – 2014

SERVICE & LEADERSHIP

Graduate Student Representative, Graduate Studies Committee

Department of Mathematics, Michigan State University

2025 – 2026

East Lansing, MI, US

- Represented all mathematics graduate students at committee meetings with faculty.
- Planned and prepared Responsible Conduct of Research (RCR) workshops with the Graduate Chair.
- Met with prospective graduate students and hosted graduate student town hall meetings on the state of the department.
- Served as a point of contact for graduate students with grievances and directed them to department and university resources.

President / Board Member

Association for Women in Mathematics (AWM), MSU Chapter

August 2022 – August 2025

East Lansing, MI, US

- Organized seminars, career workshops, informational sessions on academic life and social events for women in mathematics.

Organizing Committee Member

Graduate Student Topology and Geometry Conference, Michigan State University

April 2024

East Lansing, MI, US

- Handled communication with speakers and attendees, room assignments and talk scheduling, and moderated sessions.

President

METU Math Club

February 2015 – December 2018

Ankara, Türkiye

- Organized seminars, workshops and regular reading groups in topology, algebra and other areas.

CONFERENCES & WORKSHOPS ATTENDED

Agentic AI (hot topic course), AIDMM NSF Research Traineeship, Michigan State University	Fall 2026
AI, Data and PDE Workshop, Michigan State University	April 2025
Graduate Student Topology and Geometry Conference, Michigan State University	April 2024
Mid-Atlantic Topology Conference, Northeastern University	March 2024
SIAM Great Lakes Section Meeting, Michigan State University	October 2023
Computational Persistence, Purdue University	September 2023
Randomness in Topology and its Applications, IMSI, Chicago	March 2023
LMS Workshop: Applied Algebraic Topology, Queen Mary University of London (online)	February 2022
Topological Insights in Neuroscience, IMSI (online)	May 2021
Topological Data Analysis, IMSI (online)	April 2021

TECHNICAL SKILLS**Programming:** Python, R, SQL**Methods:** Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation**Libraries:** PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM**Tools:** Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, L^AT_EX**Domain Knowledge:** Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis**REFERENCES**

Available upon request.